

Specific rigidity factors in simple unimolecular bond fission reactions

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Specific rate constants, $k(E, J)$, and thermally averaged rate constants for simple unimolecular bond fission reactions and the corresponding combination reactions to a first approximation are represented by phase space theory (PST) which neglects the anisotropy of the potential-energy surface. A central quantity in this treatment is the number of open reaction channels, $W(E, J)$, which in PST can be expressed in approximate analytical form. Accounting for the anisotropy of the potential introduces additional specific rigidity factors, $f_{\text{rigid}}(E, J)$, being between zero and unity, which have to be multiplied by $W(E, J)$ from PST. After thermal averaging, thermal rigidity factors, $f_{\text{rigid}}(T)$, are obtained. This article shows examples of specific rigidity factors as derived from the statistical adiabatic channel model (SACM) for selected anisotropies of the potential.

1 Introduction

Simple unimolecular bond fission reactions and the reverse bond-forming combination processes involve loose activated complexes. Their rates to a first approximation are characterized by Gorin-type loose transition state theory (TST) and phase space theory (PST); activated complexes here are located at the top of the centrifugal barriers. It is well known that PST describes the limiting situation of an isotropic potential between the two dissociation fragments. In reality, the anisotropy of the potential results in a tightening of the activated complexes, *i.e.* their position moves towards smaller inter-fragment distances than given by the centrifugal barriers, and their energy levels move towards higher energies compared with those of the completely loose transition states. As a consequence, the rates decrease. In the spirit of the usual distinction between rigid and loose activated complexes, one may term this phenomenon ‘rigidity induced by the anisotropy of the potential.’ It appears reasonable to quantify this effect by comparing the rates with those predicted in the hypothetical absence of the anisotropy. Therefore, it was proposed to call the ratio of the real rate to the rate given by PST a rigidity factor, f_{rigid}^{1-3} .

Considering thermal rate constants, k_{∞} , for bond fission reactions or the reverse combination reactions in the high-pressure limit, one has thermal rigidity factors $f_{\text{rigid}}(T)$, defined by

$$k_{\infty} = f_{\text{rigid}}(T) k_{\infty}^{\text{PST}} \quad (1)$$

where k_{∞} is the rate constant and k_{∞}^{PST} denotes the high-pressure rate constant given by PST. Besides $f_{\text{rigid}}(T)$, one has to investigate specific rigidity factors, $f_{\text{rigid}}(E, J)$, which enter the specific rate constants $k(E, J)$ of unimolecular bond fissions. Statistical unimolecular rate theory expresses $k(E, J)$ as

$$k(E, J) = W(E, J) / h \rho(E, J) \quad (2)$$

with the number of activated complex states (or of open channels), $W(E, J)$, and the density of states, $\rho(E, J)$, at the total energy E and the total angular momentum (quantum number J) of the dissociating species. Comparing $W(E, J)$ with the corresponding expression from PST, defines $f_{\text{rigid}}(E, J)$ as

$$W(E, J) = f_{\text{rigid}}(E, J) W^{\text{PST}}(E, J) \quad (3)$$

Clearly, thermal and specific rigidity factors, $f_{\text{rigid}}(T)$ and $f_{\text{rigid}}(E, J)$, are linked. The high-pressure dissociation rate con-

stant k_{∞} in the language of TST is written as

$$k_{\infty} = \frac{kT}{h} \frac{Q^{\ddagger}}{Q} \exp\left(-\frac{E_0}{kT}\right) \quad (4)$$

with the activated complex partition function $Q^{\ddagger}$ given by

$$Q^{\ddagger} = \sum_{J=0}^{\infty} (2J+1) \int_0^{\infty} W(E, J) \exp\left(-\frac{E-E_0}{kT}\right) \frac{dE}{kT} \quad (5)$$

With the knowledge of $f_{\text{rigid}}(E, J)$ and $W^{\text{PST}}(E, J)$, one obtains $f_{\text{rigid}}(T)$ from

$$f_{\text{rigid}}(T) = Q^{\ddagger} / Q^{\ddagger \text{PST}} \quad (6)$$

$W(E, J)$ also enters the rate constants k_{bim} for complex-forming bimolecular reactions for which, in the low-pressure limit, one has

$$k_{\text{bim}} \propto \sum_{J=0}^{\infty} (2J+1) \int_0^{\infty} \left[\frac{W_a(E, J) W_b(E, J)}{W_a(E, J) + W_b(E, J)} \right] \times \exp\left(-\frac{E}{kT}\right) \frac{dE}{kT} \quad (7)$$

with the subscripts a and b specifying the activated complexes for entering and leaving the intermediate adduct of the reaction.⁴ Eqn. (7) includes the high-pressure association or capture rate constant [at $W_b(E, J) \gg W_a(E, J)$]. Knowing $W^{\text{PST}}(E, J)$ and $f_{\text{rigid}}(E, J)$ for entrance and exit of the intermediate adduct directly leads to the overall rate constant.

Using the notation of thermal and specific rigidity factors provides a number of practical advantages. Apart from being accessible by counting algorithms, partial $W^{\text{PST}}(E, J)$ for the transitional modes have been approximated well by smoothed analytical expressions;⁵⁻⁸ for simple cases, explicit rigorous relations are also available.⁸ The total $W^{\text{PST}}(E, J)$ is then obtained conveniently⁹ by convolution of the partial $W^{\text{PST}}(E, J)$ for the transitional modes with the contributions of the conserved modes. With this direct access to $W^{\text{PST}}(E, J)$, $f_{\text{rigid}}(E, J)$ provides the anisotropy correction to $W^{\text{PST}}(E, J)$. The full advantage, however, is only realized when analytical approximations are also available for $f_{\text{rigid}}(E, J)$. This is the subject of the present article. Characterizing $f_{\text{rigid}}(E, J)$, of course, requires the complete solution of the dynamic problem. For this purpose, we use the combination of classical trajectories (CT), detailed statistical adiabatic channel calculations (SACM) and simplified oscillator versions of SACM, as elaborated in ref. 10–13. To the extent that parametrized analytical

results can be obtained $f_{\text{rigid}}(E, J)$ will be expressed analytically. The given examples then serve as an orientation for practical reaction systems where a more detailed treatment can only be performed when better knowledge of the individual potentials becomes available.

2 Specific rigidity factors, $f_{\text{rigid}}(E, J)$, for ion–permanent dipole potentials

The anisotropy of an ion–permanent dipole interaction potential provides the simplest illustration of specific rigidity factors. We consider the potential

$$V(r, \gamma) = -q\mu_D \cos \gamma / r^2 + l(l+1)\hbar^2 / 2\mu r^2 \quad (8)$$

(r = distance between the centres of mass of the point charge and the linear dipole rotor, γ = angle between the rotor axis and the line between the centres of mass, q = ion charge, μ_D = dipole moment of the rotor, μ = reduced mass and l = quantum number of orbital motion). We can assure that $l \approx J$ when J is not too small. (For small J , a more appropriate relation between l and J is given, to a good approximation, by $l(l+1) = j(j+1) - 2m^2 + J(J+1)$ with the rotor quantum numbers j and m).¹⁰ Adding the r -dependent eigenvalues of the linear rotor in the angular potential to the maximum attraction potential, *i.e.* to eqn. (8) with $\gamma = 0$, defines the adiabatic channel potential curves $V_i(r)$. The relevant eigenvalues have been derived by matrix diagonalisation and analytical approximation in ref. 11. We use these adiabatic channel potential curves in the present work, determine the channel-specific threshold energies E_{0i} , and count the number of channels with $E_{0i} \leq E$ which defines $W(E, J)$. For details, the reader is referred to ref. 11.

Fig. 1 shows $W(E, J)$ for an $\text{H}_3^+ - \text{HCl}$ ion–dipole pair ($2\mu q\mu_D/\hbar^2 = 4442$, bound adducts only existing for $J \leq 66$; the parameters are given in the Appendix). The figure, for three J -values, compares $W^{\text{PST}}(E, J)$ with $W(E, J)$. $W^{\text{PST}}(E, J)$ corresponds to the levels of a two-dimensional linear rotor (rotational constant B in energy units) which, in classical

smoothed form, is given by

$$W^{\text{PST}}(E, J) = E/B \quad (9)$$

as shown in Fig. 1. Eqn. 9 applies when $0 \leq G \leq 1$ where G is defined by

$$G = J(J+1)\hbar^2 / 2\mu q\mu_D \quad (10)$$

for $G > 1$, one has $W^{\text{PST}}(E, J) = 0$. In Fig. 1 one observes that $W(E, J)$, for $J = 0$, is near $W^{\text{PST}}(E, J)$ whereas, for increasing J , it falls below $W^{\text{PST}}(E, J)$. One can smooth the step functions in Fig. 1 and derive specific rigidity factors $f_{\text{rigid}}(E, J)$. One obtains the important result that, in this case, there is no energy dependence of $f_{\text{rigid}}(E, J)$; however, a marked J dependence is observed as a consequence of the anisotropy of the potential. Fig. 2 illustrates the J dependence of $W(E, J)$ over larger ranges of E/B , using the real $\text{N}^+ - \text{HCN}$ as a representative example ($2\mu q\mu_D/\hbar^2 = 39432$, parameters given in the Appendix). Here, the step functions can be smoothed more easily and a larger range of J values is relevant.

The J dependence of $f_{\text{rigid}}(E, J)$ can be expressed in a general analytical form. We have performed $W(E, J)$ calculations for a series of representative systems characterized by the molecular parameters of the $(\text{H}_3^+, \text{N}^+) + (\text{HCl}, \text{CS}, \text{HCN})$ systems without polarizability contributions (parameters given in the Appendix). The results of the smoothed $W(E, J)$, except for $G < 0.2$, correspond very well to the expression

$$f_{\text{rigid}}(E, J) = 1 - G \quad (11)$$

with G defined by eqn. (10). Fig. 3 illustrates this result. For small J , there is some minor additional rigidity which in ref. 11 was approximated by using $f_{\text{rigid}}(E, J) \approx 0.94(1 - G)$ instead of eqn. (11). The present more systematic calculations showed that an empirical expression

$$f_{\text{rigid}}(E, J) \approx 1 - G - 0.064 \exp(-9G) \quad (12)$$

can better account for the additional rigidity when $G < 0.2$, see the dashed curve in Fig. 3. Eqn. (12), therefore, supersedes eqn. (21) in ref. 11. One should note that eqn. (12) applies to smoothed expressions for $W(E, J)$; the step nature of the quantum curves of $W(E, J)$ may not be well represented by eqn. (12) when high-energy resolution is required.

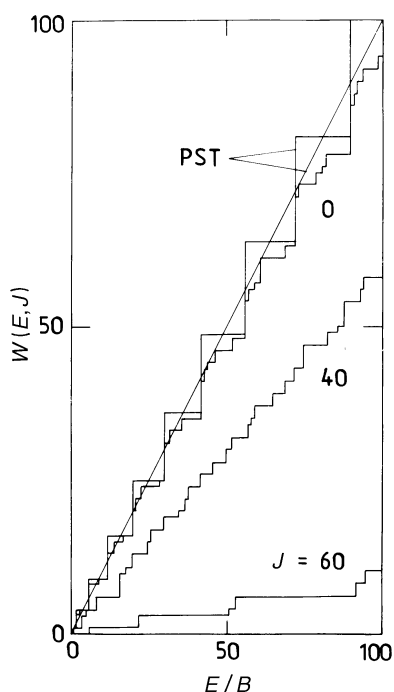


Fig. 1 Number of open channels $W(E, J)$ for the $\text{HCl} - \text{H}_3^+$ system [irregular step curves: SACM results; regular step curve at the top and smoothed curve $W(E, J) = E/B$: PST, see eqn. (9)]

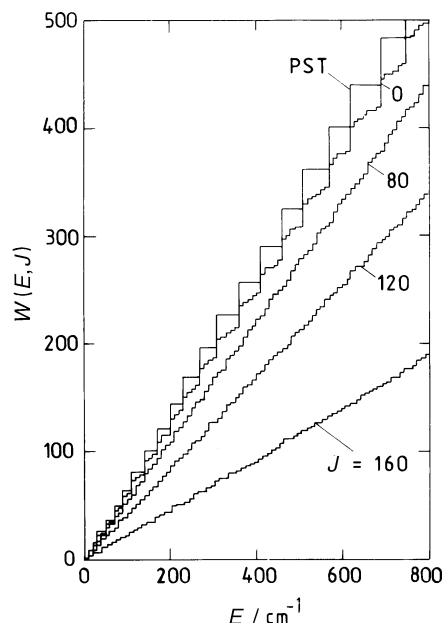


Fig. 2 Number of open channels $W(E, J)$ for the $\text{HCN} - \text{N}^+$ system (irregular step curves: SACM results, regular step curve at the top: PST)

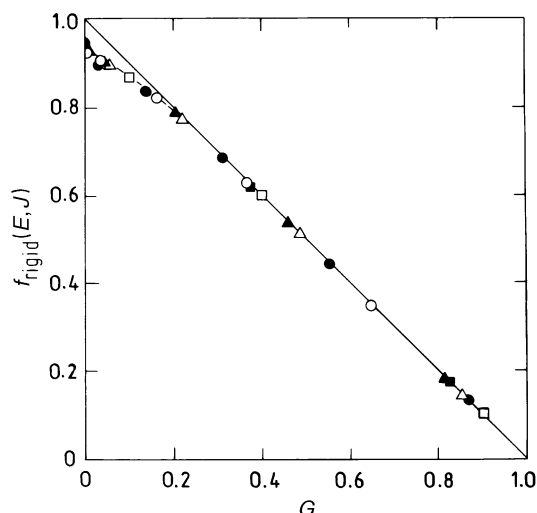


Fig. 3 Specific rigidity factors, $f_{\text{rigid}}(E, J)$, for ion–permanent dipole systems [$G = J(J+1)\hbar^2/2\mu q\mu_D$; (○) HCN + (H_3^+ , N^+); (□) HCl + (H_3^+ , N^+); (△) CS + (H_3^+ , N^+); open symbols: H_3^+ ; filled symbols: N^+ ; full line: eqn. (11); dashed line: eqn. (12)]

Before considering other anisotropies of the potential, we briefly illustrate the relation between thermal and specific rigidity factors. Using eqn. (5), (6) and (11), leads to

$$f_{\text{rigid}}(T) = \left[\sum_{J=0}^{J_{\text{max}}} (2J+1)(1-G) \right] / \left[\sum_{J=0}^{J_{\text{max}}} (2J+1) \right] \quad (13)$$

where J_{max} in eqn. (10) corresponds to $G = 1$. Integrating eqn. (13) gives

$$f_{\text{rigid}}(T) = 1/2 \quad (14)$$

which agrees with the result for ion–permanent dipole capture in the classical high-temperature limit.^{11,12} With decreasing temperature, when smaller G values become more important, a decrease in $f_{\text{rigid}}(T)$ is apparent. In addition, quantum effects arise as quantified in ref. 11.

Whereas only contributions from the transitional modes determine the relation between $f_{\text{rigid}}(E, J)$ and $f_{\text{rigid}}(T)$ in the present calculation of capture rates, the implementation of $f_{\text{rigid}}(E, J)$ into bimolecular rate constants, see eqn. (7), is less direct. In this situation entrance and exit activated complexes generally have different threshold energies on the potential. In addition the expressions for $f_{\text{rigid}}(E, J)$ together with $W^{\text{PST}}(E, J)$ have to be convoluted with contributions from conserved modes; eqn. (7) then in general has to be evaluated numerically. However, having analytical expressions for $W^{\text{PST}}(E, J)$ and $f_{\text{rigid}}(E, J)$ for the transitional modes greatly facilitates this task.

3 Specific rigidity factors, $f_{\text{rigid}}(E, J)$, for ion–permanent + induced dipole potentials

The specific rigidity factors, $f_{\text{rigid}}(E, J)$, can be complicated functions of the energy and the angular momentum. We illustrate this by the more complicated mixed anisotropy of an ion–permanent + induced dipole potential. We consider

$$V(r, \gamma) = -q\mu_D \cos \gamma / r^2 + l(l+1)\hbar^2/2\mu r^2 - \alpha q^2/2r^4 \quad (15)$$

where α is the isotropic polarizability. We use the same molecular systems as in the previous section, with the addition of real or artificially varied polarizability terms. The anisotropy of the potential of eqn. (15) is non-uniform, *i.e.* at small r the potential is less anisotropic than at large r . As a consequence one expects specific rigidity factors which, with increasing energy, increase over those of Section 2. Fig. 4 and 5 for the $\text{H}_3^+ - \text{HCl}$ and $\text{H}_3^+ - \text{HCN}$ systems illustrate the effect of increasing α .

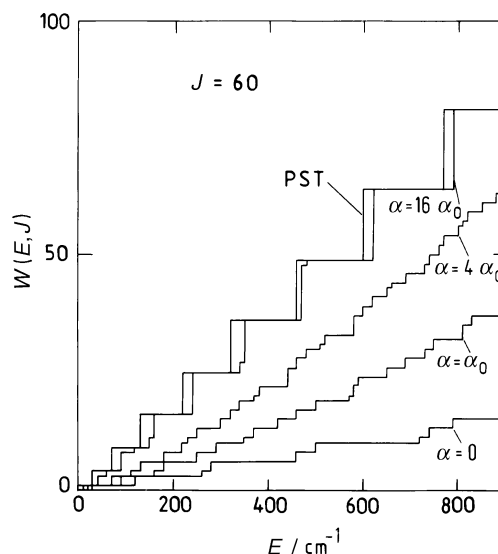


Fig. 4 Number of open channels, $W(E, J)$, for ion–permanent + induced dipole systems (SACM calculations for the HCl + H_3^+ system with varied polarizability α , $\alpha_0 = 2.60 \times 10^{-24} \text{ cm}^3$, $J = 60$)

Before determining explicit specific rigidity factors, we shall inspect the J dependence of the reaction threshold energy $E_0(J)$. In PST, one has $E_0(J) = 0$ for $G = J(J+1)\hbar^2/2\mu q\mu_D \leq 1$ and

$$E_0(J) = (G-1)^2 \mu_D^2 / 2\alpha \quad (16)$$

for $G \geq 1$.¹⁴ SACM¹⁰ at small G leads to

$$E_0(J) \approx 3BG^2/2(1 + \alpha B/3\mu_D^2) \quad (17)$$

for larger G , $E_0(J)$ at first tends towards the expression given by an SACM oscillator model

$$E_0(J)/B \approx [2(1-G)]^{-1} - 1/2 \quad (18)$$

before $E_0(J)$, finally approaches eqn. (16). The real $E_0(J)$ has to be calculated numerically as described in ref. 11. $E_0(J) > 0$ for

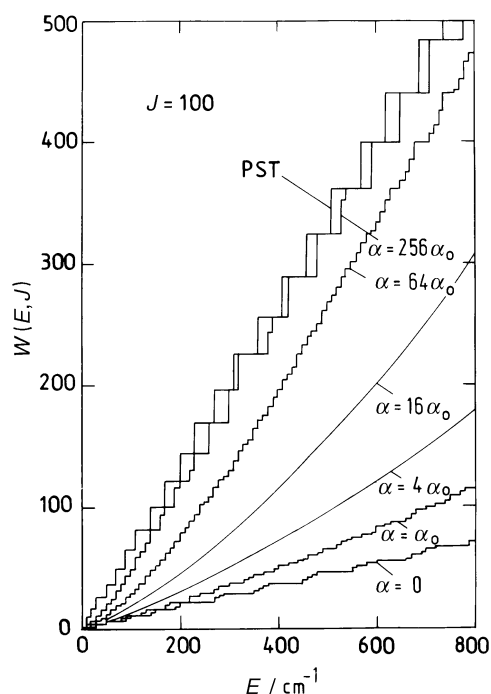


Fig. 5 As Fig. 4, for the HCN + H_3^+ system ($\alpha_0 = 2.56 \times 10^{-24} \text{ cm}^3$, $J = 100$; the curves for $\alpha = 4\alpha_0$ and $16\alpha_0$ have been smoothed)

$0 < G < 1$ is already observed in the ion-permanent dipole case of the previous section, but the small effect was neglected in the discussion of the specific rigidity factors in Section 2. Nevertheless, its influence is important in the quantum range of the capture rate constants for $kT \leq B$, see details in ref. 11. Complex-forming bimolecular reactions, eqn. (7), may be particularly sensitive to different $E_{0a}(J)$ and $E_{0b}(J)$ for the entrance and exit activated complexes (e.g. ref. 15). Then, the lowest adiabatic channels with their adiabatic zero-point energies may have to be treated explicitly with the approximate channel potential of ref. 11.

In order to facilitate applications, in the present work we performed explicit SACM calculations for the lowest channel, determining $E_0(J)$. The induced dipole was characterized by the reduced parameter¹²

$$C = \alpha B / 2\mu_D^2 \quad (19)$$

The threshold energy for $G = J(J+1)\hbar^2/2\mu q\mu_D = 1$ can be expressed by this parameter through

$$E_0(G=1)/B = 0.62C^{-0.35} \quad (20)$$

For $G \leq 1$, eqn. (20) is approached by

$$E_0(G)/B \approx \frac{\{[2(1-G)]^{-1} - 0.5\}0.62C^{-0.35}}{\{[2(1-G)]^{-1} - 0.5\} + 0.62C^{-0.35}} \quad (21)$$

At $G > 1$, eqn. (20) is approached by

$$E_0(G \gg 1)/B = (G-1)^2/4C$$

the threshold energy obtained from eqn. (20) through

$$E_0(G)/B = (G-1)^2/4C + 0.62C^{-0.35}[1 + 6.6(G-1)^{0.67}] \quad (22)$$

In Fig. 6, for several values of C , we demonstrate the quality of our analytical results by comparing eqn. (21) and (22) with accurate $E_0(G)$ (the SACM calculations were performed for the HCN + N⁺ system with arbitrarily varied α).

We now investigate whether, for the ion-permanent + induced dipole case, an approximate general expression for the specific rigidity factors can also be derived. One has to modify eqn. (12) by an energy dependence which leads $f_{\text{rigid}}(E, J)$ from the low-energy limit of eqn. (12) to the high-energy limit of unity. Furthermore, one has to account

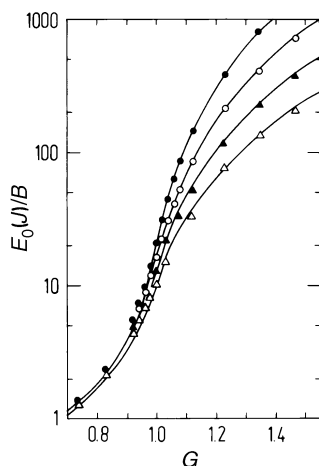


Fig. 6 Reaction threshold energies, $E_0(J)$, for ion-permanent + induced dipole potentials [$G = J(J+1)/J_0(J_0+1)$ with $J_0(J_0+1) = 2\mu q\mu_D/\hbar^2$; points: rigorous SACM calculations for the lowest channel of the HCN + N⁺ system¹¹ with $C = C_0$ (●), $2C_0$ (○), $4C_0$ (▲), $8C_0$ (△), $C_0 = \alpha B/2\mu_D^2 = 4.232 \times 10^{-5}$ and $J_0(J_0+1) = 39432$; lines: approximate representation by eqn. (21) and (22)]

for increases in $f_{\text{rigid}}(E, J)$ with increasing α or, in reduced form, with increasing C .

The stepped nature of $W(E, J)$ slightly complicates the representation of $f_{\text{rigid}}(E, J)$. Here, we have only smoothed $W^{\text{PST}}(E, J)$, by employing $W^{\text{PST}}(E, J) = E/B$, without smoothing $W(E, J)$. Besides the J dependence, the E dependence of $f_{\text{rigid}}(E, J)$ also has to be represented for the mixed ion-permanent + induced dipole system. Fig. 7 illustrates this E -dependence for the HCN + N⁺ system, arbitrarily varying the polarizability parameter C from eqn. (19). For small C , $f_{\text{rigid}}(E, J)$ rises slowly from the low-energy limit of the ion-permanent dipole system; for large C , the limit $f_{\text{rigid}}(E, J) \approx 1$ from PST corresponding to the isotropic polarizability potential is rapidly approached. It has been shown¹² that the system can be completely characterized by the two parameters $J_0(J_0+1) = 2\mu q\mu_D/\hbar^2$ and $C = \alpha B/2\mu_D^2$; this was empirically confirmed in the present work. We, therefore, represent the results of Fig. 7 using these two parameters. In the simplest way $f_{\text{rigid}}(E, J)$ is expressed relative to the value $f_{\text{rigid}}^0(E, J)$ from the ion-permanent dipole system, eqn. (12). With

$$E^* = E - E_0(J) \quad (23)$$

one has $f_{\text{rigid}}(E^*, J) > 0$ for $E^* \geq 0$ and $f_{\text{rigid}}^0 = 0$ for $E^* < 0$. In Fig. 8 we have plotted in reduced form the deviation of $f_{\text{rigid}}(E^*, J)$ from $f_{\text{rigid}}^0(E^*, J)$. We have chosen the variables

$$Y = \frac{1 - f_{\text{rigid}}(E^*, J)}{1 - f_{\text{rigid}}^0(E^*, J)} \quad (24)$$

and

$$X = E^*C/BG \quad (25)$$

For a given G , calculations with different C fall on one curve; the HCN + N⁺ and HCN + H₃⁺ systems were chosen as

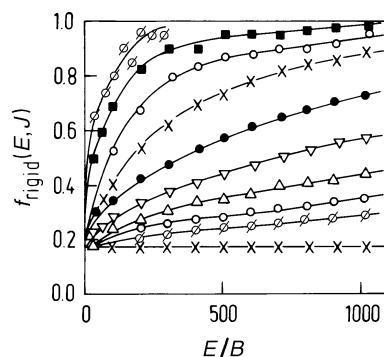


Fig. 7 Specific rigidity factors, $f_{\text{rigid}}(E, J)$, for ion-permanent + induced dipole potentials [the system of Fig. 6 with $J = 180$, $G = 0.826$; from bottom to top $C = 0$ (×), C_0 (○), $2C_0$ (○), $4C_0$ (×), $8C_0$ (▽), $16C_0$ (●), $32C_0$ (×), $64C_0$ (○), $128C_0$ (■), $256C_0$ (△)]

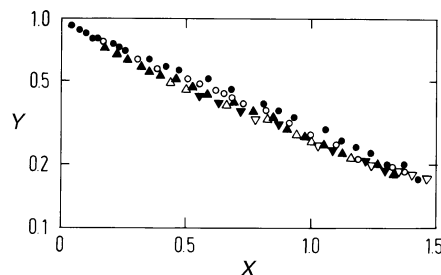


Fig. 8 Specific rigidity factors, $f_{\text{rigid}}(E, J)$, for ion-permanent + induced dipole potentials. Reduced representation of eqn. (24)–(26), see text [results for $G = 0.256$ (○), 0.368 (●), 0.653 (▲), 0.826 (△), 1.019 (▽) and 1.233 (▽); SACM calculations for the HCN + N⁺ and HCN + H₃⁺ systems].

examples, with polarizabilities arbitrarily varied from α to 256α . As with the variables Y and X , the results for different G , covering values from below to above unity, fall practically on a single curve [the minor G -trend in practice can be neglected, being smaller than the irregularities from the step character of $W(E, J)$]. Using the parameters in eqn. (23)–(25), we obtain an analytical representation for the rigidity factors $f_{\text{rigid}}(E, J)$ even in this complicated case of mixed anisotropy, the results of Fig. 8 being approximated by the relation

$$Y \approx \exp(-1.31X) \quad (26)$$

The quality of eqn. (21)–(26) could be controlled by integrating the smoothed $W^{\text{PST}}(E, J)f_{\text{rigid}}(E, J)$ over a thermal distribution in E, J and comparing the derived capture rate constant with the analytical results from ref. 11 and 12 (see also the slightly less accurate expressions in ref. 16).

4 Specific rigidity factors, $f_{\text{rigid}}(E, J)$, for atom–linear rotor valence potentials

The concept of this article can be applied equally well to other types of potential. Interesting models, which we have treated by a combination of SACM and CT calculations, are atom–linear rotor¹⁷ and linear rotor–linear rotor systems¹⁸ with valence potentials. For simplicity, the anisotropies were assumed to be of the ion–dipole type (*i.e.* of $\cos \gamma$ form) or of the dipole–dipole type. However, the anisotropy amplitudes were linked empirically¹⁹ to Morse-type attraction potentials. In the following, we illustrate the results by elaborating analytical expressions for $f_{\text{rigid}}(E, J)$ in the atom–linear rotor system. The more complicated linear rotor–linear rotor case will be treated elsewhere.¹⁸

Instead of the potentials in eqn. (8) and (15) we now consider a simplified valence potential

$$V(R, \gamma) = -2D \exp[-\beta(r - r_e)] + D \exp[-2\beta(r - r_e)] + V_{\text{oe}}[(1 - \cos \gamma)/2] \exp[-\beta(r - r_e)] \quad (27)$$

with the Morse parameter β , the equilibrium value r_e of the centre-of-mass distance r , the Morse dissociation energy D and the anisotropy amplitude V_{oe} . The latter is related to the r -dependent quantum $\varepsilon(r)$ of the doubly degenerate bending vibration of the atom–linear rotor adduct. Eqn. (27) corresponds to

$$\varepsilon(r) = \varepsilon(r_e) \exp[-\beta(r - r_e)/2] \quad (28)$$

and

$$V_{\text{oe}} \approx [\varepsilon(r_e)]^2/B \quad (29)$$

According to the analysis in ref. 17, the parameters βr_e , kT/D , and V_{oe}/D completely characterize thermal capture rate constants. By analogy, we look for expressions for $f_{\text{rigid}}(E, J)$ as a function of βr_e , B/D , and

$$C' = [\varepsilon(r_e)]^2/B D \quad (30)$$

For actual calculations, we have started from the O + NO system (for parameters see Appendix) arbitrarily varying its properties in order to investigate the effects of varying details of the potential. First, we consider a model with real NO₂ parameters, a potential $V(r, \gamma)$ given by eqn. (27) and a quasi-diatomic centrifugal energy $l(l+1)\hbar^2/2\mu r^2$ corresponding to eqn. (8) and (15). Again we adopt $l \approx J$ when J is not too small, see Section 2. Fig. 9 shows SACM calculations of $W(E, J)$ in comparison with $W^{\text{PST}}(E, J)$. Adiabatic channel eigenvalues in approximate analytical form from the ion–dipole study¹¹ were adopted by replacing the r -dependent anisotropy amplitudes $2q\mu_b/r^2$ from eqn. (8) by $V_{\text{oe}} \exp[-\beta(r - r_e)]$ from eqn. (27). Fig. 9 shows that this system with its uniform anisotropy character, as Fig. 1 and 2, leads to an energy-

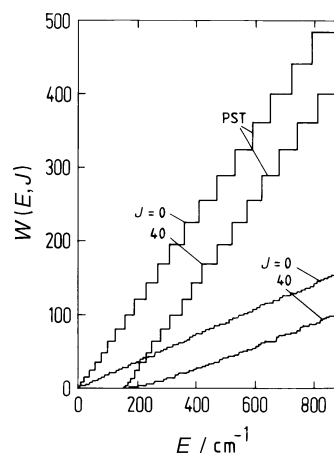


Fig. 9 Number of open channels, $W(E, J)$, for an NO₂ valence potential with the model anisotropy of eqn. (27)–(29) [PST: $V_{\text{oe}} = 0$ in eqn. (27); for molecular parameters, see the Appendix]

independent specific rigidity factor for $J = 0$. [One should emphasize that at energies very close to the threshold energy the stepped nature of $W(E, J)$ prevents a meaningful definition of $f_{\text{rigid}}(E, J)$, *i.e.* $W(E, J)$ coincides with $W^{\text{PST}}(E, J)$ to some extent.] For $J > 0$, an energy dependence of $f_{\text{rigid}}(E, J)$ becomes apparent. With increasing J , $f_{\text{rigid}}(E, J)$ near the threshold energy, $E_0(J)$, starts at lower values; however, at high energies, $f_{\text{rigid}}(E, J = 0)$ is approached. Fig. 10 demonstrates this behaviour.

If there were no E or J dependence of $f_{\text{rigid}}(E, J)$, one would have

$$f_{\text{rigid}}(T) \approx f_{\text{rigid}}(E, J) \quad (31)$$

such that the corresponding analytical expressions for $f_{\text{rigid}}(T)$ could be used, *i.e.*

$$f_{\text{rigid}}(T) \approx 2.85BD/[\varepsilon(r_e)]^2 = 2.85/C' \quad (32)$$

(being valid for $f_{\text{rigid}} < 0.8$). Eqn. (32) with $C' = 12.9$ for NO₂ gives $f_{\text{rigid}}(T) = 0.22$ which is somewhat below the results from Fig. 9 [$f_{\text{rigid}}(E, J = 0) = 0.31$]. The difference between the two values is due to the low-energy decrease of $f_{\text{rigid}}(E, J)$ with increasing J ; furthermore, one has $E_0(J) > E_0^{\text{PST}}(J)$, which, however, is only of minor importance.

In the following, the specific rigidity factors are characterized in more detail. First, we briefly consider $E_0(J)$. As $E_0^{\text{PST}}(J)$ is easily derived numerically as the maximum of eqn. (27) with $V_{\text{oe}} = 0$ and an added centrifugal energy $E_{\text{cent}}(r) = l(l+1)\hbar^2/2\mu r^2$, we do not elaborate here empirical relations between $E_0^{\text{PST}}(J)$ and molecular parameters. We just recall that

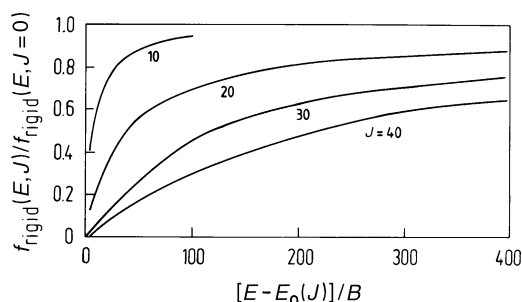


Fig. 10 Specific rigidity factors, $f_{\text{rigid}}(E, J)$, for an NO₂ valence potential [smoothed curves of $W(E, J)$; $f_{\text{rigid}}(E, J = 0) = 0.31$ from Fig. 9]

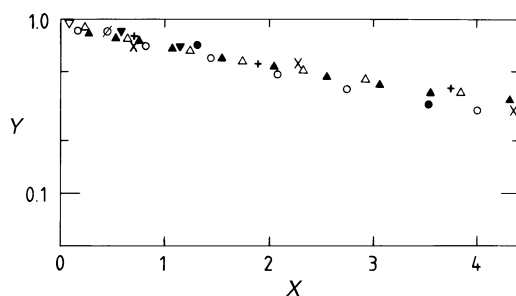


Fig. 11 Representation of specific rigidity factors, $f_{\text{rigid}}(E, J)$, for valence potentials by eqn. (36)–(38) [NO_2 with $C' = 12.9$; $J = 10$ (●), 20 (○), 30 (▲), 40 (△), 60 (▼), 80 (△); $C' = 4 \times 12.9$: $J = 20$ (▼), 40 (△), 60 (◇), see text]

the J dependence of $E_0^{\text{PST}}(J)$ can often be represented by²⁰

$$E_0^{\text{PST}}(J) \approx C_v [J(J+1)]^\nu \quad (33)$$

with $1 \leq \nu \leq 1.5$. For the actual $E_0(J)$, one finds

$$E_0(J) - E_0^{\text{PST}}(J) \approx E_0(J=0) + D_v [J(J+1)]^{\nu/2} \quad (34)$$

with $D_v \propto \sqrt{C'}$. As the values of $E_0(J) - E_0^{\text{PST}}(J)$ are small for moderately large J , they are not considered further here (they cannot be neglected, however, for capture rate constants at $kT < B$).

Next, we represent the high-energy limits of the specific rigidity factors. Analogous to eqn. (32) with C' from eqn. (30) we find

$$f_{\text{rigid}}(\text{large } E, J) \approx 4.0/C' \quad (35)$$

as long as $f_{\text{rigid}}(E, J) < 0.8$ [the approach of $f_{\text{rigid}}(E, J) \rightarrow 1$ for $C' \rightarrow 0$ could be expressed in a manner analogous to the corresponding $f_{\text{rigid}}(T)$].¹⁷ We have verified eqn. (35) by arbitrarily varying $\epsilon(r_e)$ for the NO_2 system.

Finally, a representation of the J dependence of $f_{\text{rigid}}(E, J)$ is given. Analogous to eqn. (26), an exponential approach to the high-energy limit is found, however, with different definitions of Y and X . We now express

$$Y = 1 - f_{\text{rigid}}(E, J)/f_{\text{rigid}}(\text{large } E, J) \quad (36)$$

as a function of

$$X = [E - E_0(J)]/E_0^{\text{PST}}(J) \quad (37)$$

For the NO_2 system {with $E_0^{\text{PST}}(J) \approx 0.276 [J(J+1)]^{1.17} \text{ cm}^{-1}$ } and arbitrarily varied C' one obtains the results of Fig. 11 which are represented by

$$Y = \exp(-0.285X) \quad (38)$$

Apparently in this case a C' dependence only enters *via* $f_{\text{rigid}}(\text{large } E, J)$ from eqn. (35). More systematic studies of additional influences of molecular parameters are required.

5 Influence of long-range potentials and finite rotor dimensions on $f_{\text{rigid}}(E, J)$ for atom–linear rotor valence potentials

In this section, we briefly consider more specific effects which may modify $f_{\text{rigid}}(E, J)$ from Section 4 in practical applications. First, one has to consider the use of a quasi-diatomic centrifugal energy

$$E_{\text{cent}}(r) = l(l+1)\hbar^2/2\mu r^2 \quad (39)$$

The corresponding effects on $f_{\text{rigid}}(E, J)$ can be quantified conveniently by considering the rotational energy of a linear triatomic molecule.²⁰ Instead of eqn. (39) (for a linear molecule ABC dissociating into AB + C with a B–C bond length of r

and an A–B bond length of r_{2e}), we use

$$E_{\text{cent}}(r) = B_e J(J+1)F(r) \quad (40)$$

with

$$F(r) = \frac{m_C m_B r_e^2 + m_A m_B r_{2e}^2 + m_A m_C (r_e + r_{2e})^2}{m_C m_B r^2 + m_A m_B r_{2e}^2 + m_A m_C (r + r_{2e})^2} \quad (41)$$

B_e denotes the rotational constant of the linear molecule ABC at $r = r_e$. For a quasi-diatomic system ($r_{2e} = 0$), one would have

$$F(r) = (r_e/r)^2 \quad (42)$$

and $B_e = \hbar^2/2\mu r_e^2$, i.e. eqn. (40) and (39) are the same.

Changing the centrifugal energy has a marked effect on the threshold energies $E_0^{\text{PST}}(J)$ and $E_0(J)$. Fig. 12 illustrates the effect which is relatively more important for larger J whereas there is no effect for $E_0(J = 0)$. The decrease in $E_0(J)$ upon using the centrifugal potential of eqn. (40) leads to an increase in the thermal capture rate constants. However, neither $f_{\text{rigid}}(\text{large } E, J)$ from eqn. (35) nor the energy dependence of $f_{\text{rigid}}(E, J)$ given by eqn. (36)–(38) are influenced to a noticeable extent. The effect, therefore, can be accounted for easily on the PST level by calculating $E_0^{\text{PST}}(J)$ with the centrifugal energy of eqn. (40).

In practical applications to real molecules the simple valence potential model of eqn. (27) may not be sufficient. Little is known about the transition between short-range valence and long-range electrostatic interaction potentials. In some cases, there may be an intermediate valence isomer such as HOHO between HOOH and two HO dipoles.²¹ Likewise, there may be an OON valence isomer between ONO and an NO dipole and an O quadrupole.²² Modifying the Morse potential in eqn. (27) modifies $E_0^{\text{PST}}(J)$ and, hence, the thermal capture rate constants. We have considered a long-range potential for NO_2 with the properties estimated in ref. 22 (see Appendix). The outer potential had an anisotropy analogous to the inner one, i.e. of the type in eqn. (27)–(29) with the parameters given in the Appendix. We obtained $E_0^{\text{PST}}(J)$ as shown in Fig. 12.

$E_0^{\text{PST}}(J)$ decreases further compared with the simple Morse valence potential which leads to increases in the capture rate constants. However, as long as the anisotropy amplitude, $V_0(r)$, is linked to the maximum attraction potential, $V(r)$, through

$$V_0(r) = \{[\epsilon(r_e)]^2/B\} \{1 - \sqrt{1 - |V(r)|/D}\} \quad (43)$$

which includes eqn. (27)–(29) for a simple Morse valence potential, the uniform anisotropy of the potential generates

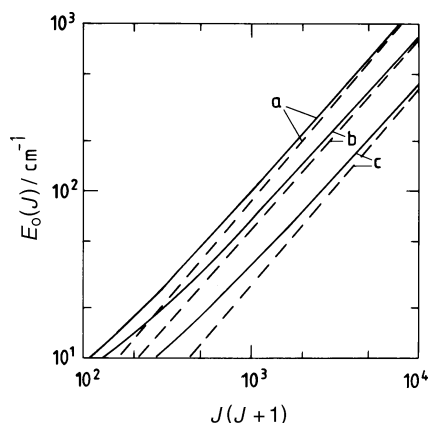


Fig. 12 Threshold energies, $E_0(J)$, for bond fission of NO_2 (full lines: SACM; dashed lines: PST; a: simple valence potential with quasi-diatomic centrifugal energy; b: as a, but true centrifugal energy of eqn. (40); c: as b, but with long-range potential, see the Appendix)

the same specific rigidity factors as before, *i.e.* eqn. (30), (35)–(38) remain valid.

Non-uniform anisotropy changes the situation. The long-range electrostatic potential generally has a smaller anisotropy than the valence Morse potential. As a consequence, $f_{\text{rigid}}(E, J)$ is closer to unity at small energies than at large energies, *i.e.* eqn. (35) does not hold for all energies at $J = 0$. This effect may even lead to $f_{\text{rigid}}(\text{small } E, J = 0) \approx 1$ near the threshold energy. We do not investigate these effects further, which probably characterize true NO_2 , but leave these investigations to a special study of NO_2 dissociation. One should also note that the open-shell character of the $\text{O} + \text{NO}$ fragments can have marked effects on $f_{\text{rigid}}(E, J)$, particularly near the threshold.^{11,23}

6 Conclusions

The present explicit calculations and analytical representations of specific rigidity factors, $f_{\text{rigid}}(E, J)$, allow the effects of the anisotropies of potentials on the specific rate constants of unimolecular reactions to be explored. The derived equations show in which way parameters of the potential influence the kinetics. Individual molecules may show special effects which are not accounted for by the present relations. However, the general behaviour should be well characterized because the adiabatic treatment used in the present approach was shown^{12,13,17,18} to provide an excellent characterization of the dissociation dynamics.

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Appendix

(1) Molecular parameters for capture by H_3^+ : HCl : $B = 10.588 \text{ cm}^{-1}$, $\mu_{\text{D}} = 1.109 \text{ D}$, $\alpha = 2.60 \times 10^{-24} \text{ cm}^3$, $\mu = 2.79 \text{ g mol}^{-1}$. HCN : $B = 1.478 \text{ cm}^{-1}$, $\mu_{\text{D}} = 2.98 \text{ D}$, $\alpha = 2.56 \times 10^{-24} \text{ cm}^3$, $\mu = 2.72 \text{ g mol}^{-1}$. CS : $B = 0.820 \text{ cm}^{-1}$, $\mu_{\text{D}} = 1.96 \text{ D}$, $\alpha = 4.66 \times 10^{-24} \text{ cm}^3$, $\mu = 2.83 \text{ g mol}^{-1}$. Capture by N^+ : $\mu = 10.12$ (HCl), $\mu = 9.23$ (HCN) and $\mu = 10.63$ (CS) g mol^{-1} . For references, see ref. 11.

(2) Molecular parameters for NO_2 models. (i) Simple Morse valence potential: $D = 26067 \text{ cm}^{-1}$, $B(\text{NO}) = 1.704 \text{ cm}^{-1}$, $r_e = 1.197 \text{ \AA}$, $\beta = 3.417 \text{ \AA}^{-1}$, $\epsilon_e = 758.6 \text{ cm}^{-1}$. (ii) Long-range valence potential:²² Morse potential of eqn. (27) for $r \leq 1.86 \text{ \AA}$ with $\beta = \beta_e = 3.417 \text{ \AA}^{-1}$, $\beta(r) = \beta_e - 0.639 \text{ \AA}^{-1} (r - 1.86)/\text{\AA}$ for $1.86 \leq r/\text{\AA} \leq 4.14$, $\beta = 1.96 \text{ \AA}^{-1}$ for $r \geq 4.14 \text{ \AA}$.

As long as the long-range dipole–quadrupole and quadrupole–quadrupole potential $V_{\text{LR}}(r)$ between NO and O is smaller than the Morse potential, $V_{\text{LR}}(r)$ is chosen. $V_{\text{LR}}(r)$ is given by $V_{\text{LR}}(r)/B(\text{NO}) = 1512/(r/\text{\AA})^4 - 22860/(r/\text{\AA})^5$.

References

- 1 C. J. Cobos and J. Troe, *J. Chem. Phys.*, 1985, **83**, 1010.
- 2 J. Troe, *Z. Phys. Chem. Neue Folge*, 1989, **61**, 209.
- 3 J. Troe, *Combust. Flame*, 1989, **78**, 59.
- 4 J. Troe, *J. Chem. Soc., Faraday Trans.*, 1994, **90**, 2303.
- 5 W. J. Chesnavich and M. T. Bowers, *J. Am. Chem. Soc.*, 1976, **98**, 8301; *J. Chem. Phys.*, 1977, **66**, 2306; 1978, **68**, 901; in *Gas Phase Ion Chemistry*, vol. I, ed. M. T. Bowers, Academic Press, New York, 1979; W. J. Chesnavich, T. Su and M. T. Bowers, *J. Chem. Phys.*, 1980, **72**, 2641.
- 6 M. Olzmann and J. Troe, *Ber. Bunsen-Ges. Phys. Chem.*, 1992, **96**, 1372; 1994, **98**, 1563.
- 7 C. E. Klots and J. Polách, *J. Phys. Chem.*, 1995, **99**, 15396.
- 8 J. Troe, *J. Chem. Phys.*, 1983, **79**, 6017.
- 9 D. C. Astholz, J. Troe and W. Wieters, *J. Chem. Phys.*, 1979, **70**, 5107 (Appendix).
- 10 J. Troe, *J. Chem. Phys.*, 1987, **87**, 2773.
- 11 J. Troe, *J. Chem. Phys.*, 1996, **105**, 6249.
- 12 A. I. Maergoiz, E. E. Nikitin, J. Troe and V. G. Ushakov, *J. Chem. Phys.*, 1996, **105**, 6263.
- 13 A. I. Maergoiz, E. E. Nikitin, J. Troe and V. G. Ushakov, *J. Chem. Phys.*, 1996, **105**, 6270, 6277.
- 14 J. Troe, *Chem. Phys. Lett.*, 1985, **122**, 425.
- 15 D. Fulle, H. F. Hamman, H. Hippler and J. Troe, *J. Chem. Phys.*, 1996, **105**, 983.
- 16 T. Su and W. J. Chesnavich, *J. Chem. Phys.*, 1982, **76**, 5183; T. Su, *J. Chem. Phys.*, 1988, **88**, 4102; 1988, **89**, 5355.
- 17 A. I. Maergoiz, E. E. Nikitin, J. Troe and V. G. Ushakov, *J. Chem. Phys.*, in the press.
- 18 A. I. Maergoiz, E. E. Nikitin, J. Troe and V. G. Ushakov, *J. Chem. Phys.*, in the press.
- 19 J. Troe, *J. Chem. Soc., Faraday Trans.*, 1991, **87**, 2299.
- 20 J. Troe, *J. Chem. Phys.*, 1981, **75**, 226.
- 21 L. B. Harding, *J. Phys. Chem.*, 1989, **93**, 8004; 1991, **95**, 8653.
- 22 L. B. Harding, personal communication.
- 23 J. Troe, *Ber. Bunsen-Ges. Phys. Chem.*, 1995, **99**, 341.

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